

Numerical Analysis-Final Exam 2024

Name _____

ERP.No _____

1 Instructions

- Attempt all questions.
- Try to limit your solutions to the space given against each question.
- You are allowed to use simple scientific calculators having no programming features. Any wifi enabled device is strictly not allowed.
- You may use answer book for rough work.
- It is an open-book and open-notes exam.
- Sharing of notes and books is strictly prohibited.

Instructor: Abdul Majid

1. The function $f(x) = \sin x + x - 1$ has root at $x \approx 0.513$. Construct two fixed point iterations $p_{n+1} = g(p_n)$, so that one converges linearly and other quadratically for p_0 sufficiently close to the root. Perform two iterations for each g.

2. Use the most appropriate formula among Lagrange, Newton-Forward/Backward to approximate $f(0.15)$ from the following table.

x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7
$f(x)$	0	0.1105	0.2443	0.4050	0.5967	0.8244	1.0933	1.4096

3. A natural cubic spline $S(x)$ on $[0, 2]$ is given by

$$S(x) = \begin{cases} S_0(x) & \text{if } 0 \leq x \leq 1, \\ S_1(x) & \text{if } 1 \leq x \leq 2, \end{cases}$$

where $S_0(x) = 1 + x^3$. Find $S_1(x)$.

4. Use an appropriate Newton-Cotes formula to evaluate $\int_0^1 \frac{1}{\ln x}$ for $n = 1$. Also, find bound on the approximate error.

5. Find the values of h , x_1 and x_2 so that the following quadrature has degree of precision 2.

$$\int_0^1 f(x)dx = h\frac{9}{4}f(x_1) + h\frac{3}{4}f(x_2) + \text{Error}.$$

6. In Five-Point Midpoint formula of numerical differentiation, find an expression for the optimal value of the step size h , which minimizes total(Truncation+round off) error in the approximation. Assume that the maximum round off error of the machine is bounded by ϵ and the absolute value of the fifth derivative of the function is bounded by $M > 0$.

7. Using a Gaussian quadrature to exactly evaluate the integral $\int_{-1}^0 (x^3 + 1)dx$.

8. How does the matrix

$$\mathbf{A} = \begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}, k > 1$$

transform unit sphere($\|\mathbf{x}\| = 1$) in both l_2 and l_∞ norms? Use this fact to find the $\|\mathbf{A}\|_2$ and $\|\mathbf{A}\|_\infty$. Also, find spectral radius of $\mathbf{A}$.

9. Find the value of parameter a that will make the following iterative scheme convergent.

$$\begin{bmatrix} x_1^{(k)} \\ x_2^{(k)} \\ x_3^{(k)} \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{a} & -\frac{1}{a} \\ -\frac{1}{a} & 0 & -\frac{1}{a} \\ -\frac{1}{a} & -\frac{1}{a} & 0 \end{bmatrix} \begin{bmatrix} x_1^{(k-1)} \\ x_2^{(k-1)} \\ x_3^{(k-1)} \end{bmatrix} + \begin{bmatrix} \frac{10}{a} \\ \frac{11}{a} \\ \frac{12}{a} \end{bmatrix}$$

10. Without performing iterations show that Jacobi method will converge for the following linear system.

$$\begin{aligned}4x_1 - x_2 + x_3 &= 3 \\-2x_1 + 5x_2 - x_3 &= 7 \\x_1 + 2x_2 + 6x_3 &= 10.\end{aligned}$$